👥 TEAM BREAKDOWN (5–6 students)

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Lab 4 using your specific repository, layout, and networking assumptions. This version **treats Local subnets as private networks** and **treats the remote server / inter-site network as a 10.10.x.x "internet/public network"** for the purposes of NAT and routing logic. It also assumes the repository hierarchy:

```
None
automation/
├── mockup-infra/
└── week02-stateless-stateful/
    └── phase1-mockup/
```

We will produce:

✅ A revised **high-fidelity lab plan** ✅ **Detailed addressing & topologies** ✅ **Cisco-themed network section** ✅ **IaC / VS Code section** ✅ **Troubleshooting and engineering reflection** ✅ A complete **README.md** ready to be committed to the repo

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🧪 LAB 4 Simulated Internet (10.10.0.0/16) & Private LANs with Stateful vs Stateless Services

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🎯 LAB GOALS

1. **Network Setup:** Create two local labs (A & B) connected via a simulated "public/internet" network (10.10.0.0/16)
2. **NAT + Routing:** Configure static routes and NAT so local traffic reaches remote services
3. **Application Deployment:** Use `mockup-infra` from your repository to deploy stateless and stateful APIs
4. **Testing:** Compare how sessions behave locally vs across the simulated internet

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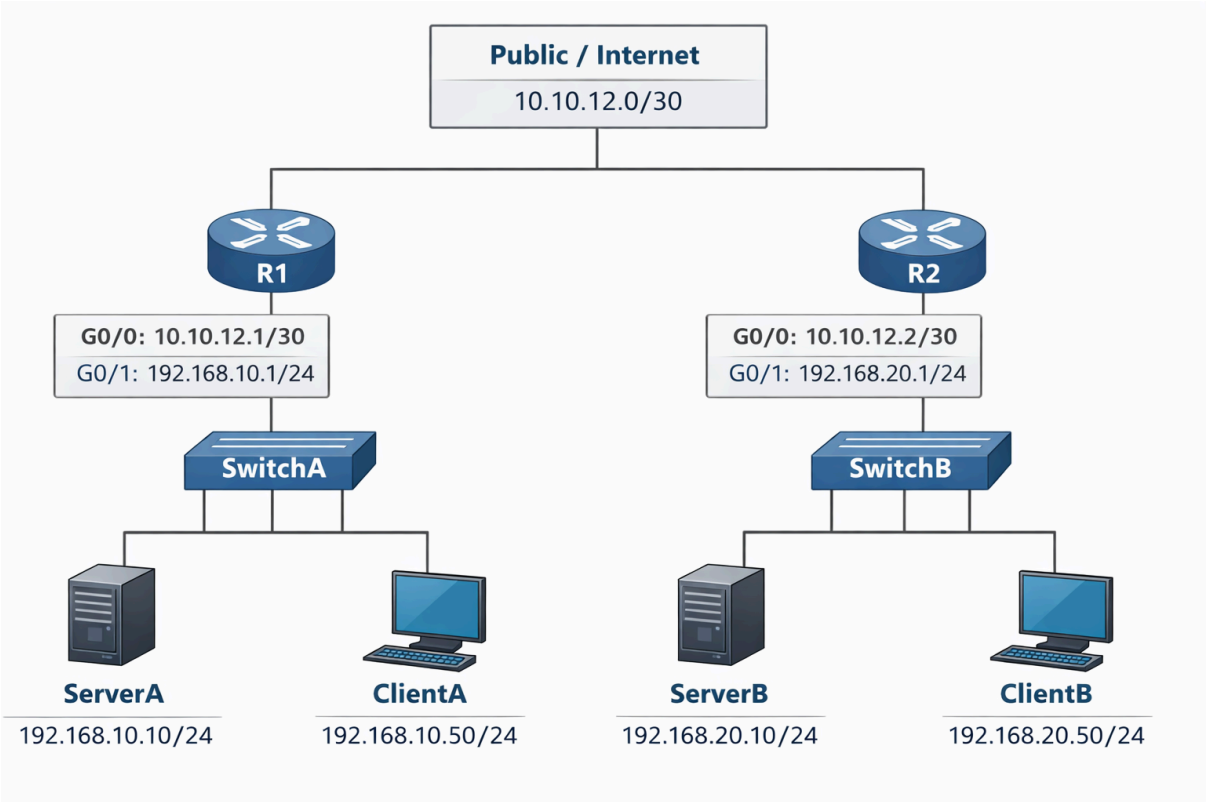
👥 TEAM BREAKDOWN (5–6 students)

Role	Count	Focus
Network Architects	2	Router configs, Cisco CLI
IaC/DevOps	2	VS Code, mockup-infra
QA/Test	1–2	Server/Client scripting, session behavior

🗺️ UPDATED NETWORK TOPOLOGY

```python
                10.10.12.0/30 "Public/Internet"
             ┌──────────────────────────┐
             |                          |
         R1-G0/0 (10.10.12.1)      R2-G0/0 (10.10.12.2)
             |                          |
   Private LAN A (192.168.10.0/24)     Private LAN B (192.168.20.0/24)
             |                          |
         SwitchA                     SwitchB
             |                          |
 ServerA (mockup)            ServerB (mockup)
 ClientA                         ClientB
```

We treat 10.10.12.0/30 as a **simulated internet** for the purposes of NAT + public reachability.



📍 ADDRESS TABLE

Component	Interface	IP	Mask	Role
R1	G0/0	10.10.12.1	/30	Public
R1	G0/1	192.168.10.1	/24	LAN A Gateway
ServerA	eth0	192.168.10.10	/24	Mockup infra A
ClientA	eth0	192.168.10.50	/24	Tester
R2	G0/0	10.10.12.2	/30	Public
R2	G0/1	192.168.20.1	/24	LAN B Gateway
ServerB	eth0	192.168.20.10	/24	Mockup infra B
ClientB	eth0	192.168.20.50	/24	Tester

🔧 PART 1 — CISCO-STYLE NETWORK CONFIGURATION

We will configure static routing + NAT to allow cross-site reachability.

🧱 R1 CONFIGURATION (LAN A)

```
None
enable
conf t

hostname R1

interface g0/0
 ip address 10.10.12.1 255.255.255.252
 ip nat outside
 no shutdown

interface g0/1
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
 no shutdown

ip route 192.168.20.0 255.255.255.0 10.10.12.2

access-list 10 permit 192.168.10.0 0.0.0.255
ip nat inside source list 10 interface g0/0 overload

end
write memory
```

```
R1#
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   10.10.12.1      YES manual up                    up
GigabitEthernet0/0/1   192.168.10.1    YES manual down                  down
GigabitEthernet0/0/2   unassigned      YES unset  down                  down
GigabitEthernet0/0/3   unassigned      YES unset  down                  down
Serial0/1/0            unassigned      YES unset  up                    up
Serial0/1/1            unassigned      YES unset  up                    up
R1#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface g0/0/1
R1(config-if)#
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)# ip nat inside
R1(config-if)# no shutdown
R1(config-if)#ip route 192.168.20.0 255.255.255.0 10.10.12.2
R1(config)#
R1(config)#access-list 10 permit 192.168.10.0 0.0.0.255
R1(config)#ip nat inside source list 10 interface g0/0 overload
                                                        ^
% Invalid input detected at '^' marker.

R1(config)#ip nat inside source list 10 interface g0/0/0 overload
R1(config)#
R1(config)#end
R1#write memory
Building configuration...
[OK]
*Feb 25 11:42:28.421: %SYS-5-CONFIG_I: Configured from console by console
```

🧱 R2 CONFIGURATION (LAN B)

```
None
enable
conf t

hostname R2

interface g0/0
 ip address 10.10.12.2 255.255.255.252
 ip nat outside
 no shutdown

interface g0/1
 ip address 192.168.20.1 255.255.255.0
 ip nat inside
 no shutdown

ip route 192.168.10.0 255.255.255.0 10.10.12.1

access-list 10 permit 192.168.20.0 0.0.0.255
ip nat inside source list 10 interface g0/0 overload

end
write memory
```

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#
R2(config)#hostname R2
R2(config)#
R2(config)#interface g0/0/0
R2(config-if)# ip address 10.10.12.2 255.255.255.252
R2(config-if)# ip nat outside
R2(config-if)# no shutdown
R2(config-if)#
R2(config-if)#interface g0/0/1
R2(config-if)# ip address 192.168.20.1 255.255.255.0
R2(config-if)# ip nat inside
R2(config-if)# no shutdown
R2(config-if)#
R2(config-if)#ip route 192.168.10.0 255.255.255.0 10.10.12.1
R2(config)#
R2(config)#access-list 10 permit 192.168.20.0 0.0.0.255
R2(config)#ip nat inside source list 10 interface g0/0/0 overload
Cannot change mapping's interface name, or the mapping already exists; remove mapping first to make change.
R2(config)#
R2(config)#end
R2#write memory
```

🧪 ROUTING & NAT VERIFICATION

On R1 & R2:

```
None
show ip route
show ip nat translations
```

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

R2#show ip nat translations
Total number of translations: 0
```

From ClientA:

```
None
ping 192.168.20.10  ← Remote server reachable
```

```
(c) Microsoft Corporation. All rights reserved.

C:\Users\CY_422012_G19>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

💻 PART 2 — IaC + MOCKUP-INFRA DEPLOYMENT (mockup-infra/README.md)

We will now set up mockup infra using the paths you specified:

```
None
automation/mockup-infra
automation/week02-stateless-stateful/phase1-mockup
```

🛠️ CLONE REPO

```
Shell
git clone https://github.com/boonsup/automation.git
cd automation
```

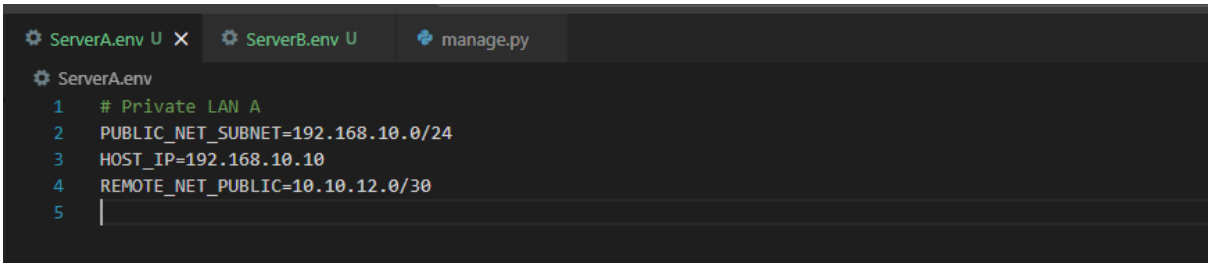
🧰 VS CODE — OPEN PROJECT

In VS Code, open the `automation` folder.

📄 CREATE SERVER-SPECIFIC ENV FILES

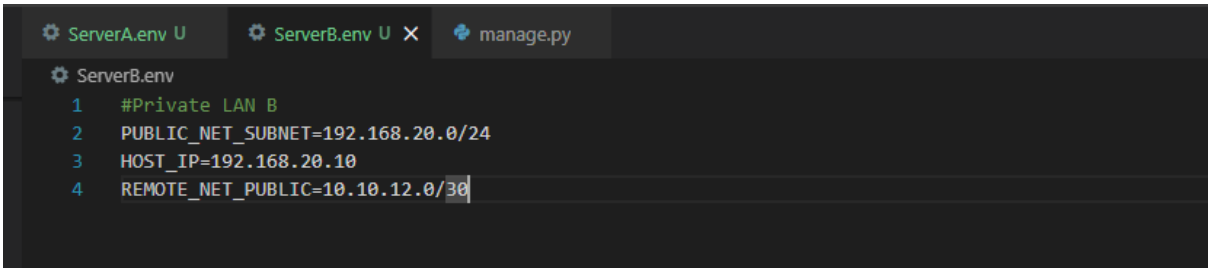
ServerA (`ServerA.env`)

```
None
# Private LAN A
PUBLIC_NET_SUBNET=192.168.10.0/24
HOST_IP=192.168.10.10
REMOTE_NET_PUBLIC=10.10.12.0/30
```



ServerB (`ServerB.env`)

```
None
# Private LAN B
PUBLIC_NET_SUBNET=192.168.20.0/24
HOST_IP=192.168.20.10
REMOTE_NET_PUBLIC=10.10.12.0/30
```



🚀 DEPLOY MOCKUP-INFRA

PowerShell (Windows Admin)

```
None
cd mockup-infra
python manage.py init
python manage.py deploy --profile week02-stateless-stateful
```

```
🔧 Initializing Mockup Infrastructure...
==============================================

🔐 [Layer 5/6 - Session] Generating TLS Certificates...
/home/sox/Documents/Network/automation/mockup-infra/manage.py:118: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future ver
n. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
  .not_valid_before(datetime.utcnow())
/home/sox/Documents/Network/automation/mockup-infra/manage.py:119: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future ver
n. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
  .not_valid_after(datetime.utcnow() + timedelta(days=365))
✅ Certificates generated successfully:
   - Private Key: /home/sox/Documents/Network/automation/mockup-infra/certs/server.key
   - Certificate: /home/sox/Documents/Network/automation/mockup-infra/certs/server.crt
   - Common Name: api.mockup.test
   - Subject Alt Names: localhost, api.mockup.test, 127.0.0.1, 172.18.0.1, 172.19.0.1

🌐 [Layer 3 - Network] Testing Isolation...
🔧 Executing: podman network inspect private_net 2>/dev/null || echo '{}'
[
    {
        "name": "private_net",
        "id": "8029c0a983f7e4f3f5e0b1689fc2ae98ee33e93e1f49de58df0db90bac79683a",
        "driver": "bridge",
        "network_interface": "podman2",
        "created": "2026-03-04T20:37:40.121546148+07:00",
        "subnets": [
            {
                "subnet": "10.89.1.0/24",
                "gateway": "10.89.1.1"
            }
        ],
        "ipv6_enabled": false,
        "internal": true,
        "dns_enabled": true,
        "ipam_options": {
            "driver": "host-local"
        }
    }
]
✅ Private network is isolated (internal: true)

✅ Infrastructure initialized successfully
```

```
🚀 Deploying Mockup Infrastructure Stack
   Bare Metal → L3 → L4 → L5/6 → L7
==================================================

🌐 [Layer 3 - Network] Testing Isolation...
🔧 Executing: podman network inspect private_net 2>/dev/null || echo '{}'
[
    {
        "name": "private_net",
        "id": "8029c0a983f7e4f3f5e0b1689fc2ae98ee33e93e1f49de58df0db90bac79683a",
        "driver": "bridge",
        "network_interface": "podman2",
        "created": "2026-03-04T20:37:40.121546148+07:00",
        "subnets": [
            {
                "subnet": "10.89.1.0/24",
                "gateway": "10.89.1.1"
            }
        ],
        "ipv6_enabled": false,
        "internal": true,
        "dns_enabled": true,
        "ipam_options": {
            "driver": "host-local"
        }
    }
]
✅ Private network is isolated (internal: true)
🔧 Executing: podman-compose up -d --remove-orphans
mockup-gateway
```

Bash (Linux)

```shell
cd mockup-infra
chmod +x scripts/*.sh
```

```
./scripts/init-env.sh
python manage.py deploy --profile week02-stateless-stateful
```

🟡 VERIFY CONTAINERS

```
None
podman ps
```

Should show:

Container	Port
stateless-api	3000
stateful-api	3001

🧪 PART 3 — TESTING STATELESS vs STATEFUL (week03-stateless-stateful/phase1-mockup/README.md)

```
ful.js
[2026-03-04T13:58:31.090Z] [INFO] Stateful server started on port 3001
```

```
less.js
[2026-03-04T13:57:39.683Z] [INFO] Stateless server started on port 3000
```

```
============================================================
🚀 Stateless Client Interactive Demo
============================================================
🔗 Connecting to stateless server...

📤 GET /health
📥 Status: 200
   Response: {
  "status": "success",
  "message": "Stateless server is healthy",
  "data": {
    "status": "healthy",
    "service": "Stateless Server",
    "timestamp": "2026-03-04T14:00:17.647Z",
    "uptime": 158.144051633,
    "memory": {
      "rss": 61075456,
      "heapTotal": 8949760,
      "heapUsed": 6932912,
      "external": 2044000,
      "arrayBuffers": 49610
    },
    "requestCount": 0,
    "serverType": "stateless",
    "note": "I have no memory of previous requests"
  },
  "timestamp": "2026-03-04T14:00:17.647Z"
}
✅ Connected to stateless server!

============================================================
🔍 Basic Stateless Requests
============================================================
📋 Testing basic server functionality...

📤 GET /health
📥 Status: 200
   Response: {
```

```
============================================================
🚀 Stateful Client Interactive Demo
============================================================
🔗 Connecting to stateful server...

📤 GET /health
📥 Status: 200
   Response: {
  "status": "success",
  "message": "Stateful server is healthy",
  "data": {
    "status": "healthy",
    "service": "Stateful Server",
    "timestamp": "2026-03-04T14:01:32.970Z",
    "uptime": 182.061147149,
    "memory": {
      "rss": 59916288,
      "heapTotal": 8949760,
      "heapUsed": 7145984,
      "external": 2044000,
      "arrayBuffers": 49610
    },
    "activeSessions": 0,
    "activeCarts": 0,
    "serverType": "stateful",
    "note": "I maintain client state across requests"
  },
  "timestamp": "2026-03-04T14:01:32.970Z"
}
✅ Connected to stateful server!

============================================================
🔄 Session Lifecycle Demonstration
============================================================
📋 Testing session management...

📤 POST /session
   Body: {
  "userId": "user001",
  "data": {
    "userAgent": "Stateful Demo Client",
    "preferences": {
      "theme": "light"
    }
```

```
================================================================================
🚀 Stateless vs Stateful Comparison Demo
================================================================================
🔄 Starting comprehensive comparison...

================================================================================
🔍 Server Health Check
================================================================================
🔵 Checking stateless server...

📤 GET /health
📥 Status: 200
   Response: {
  "status": "success",
  "message": "Stateless server is healthy",
  "data": {
    "status": "healthy",
    "service": "Stateless Server",
    "timestamp": "2026-03-04T14:02:32.148Z",
    "uptime": 292.645039805,
    "memory": {
      "rss": 63766528,
      "heapTotal": 10260480,
      "heapUsed": 8855376,
      "external": 2052342,
      "arrayBuffers": 57802
    },
    "requestCount": 4,
    "serverType": "stateless",
    "note": "I have no memory of previous requests"
  },
  "timestamp": "2026-03-04T14:02:32.148Z"
}
✅ Stateless server is healthy!
🟢 Checking stateful server...

📤 GET /health
📥 Status: 200
   Response: {
```

⚫ LOCAL TESTS (Private LAN)

From ClientA:

```shell
curl http://192.168.10.10:3000/info
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                                                    powershell

e9bb24df1475  localhost/stateful-api:latest          node start-statef...  About a minute ago  Up About a minute  0.0.0.0:3001->3001/tcp                stateful-app
PS C:\network\automation\week02-stateless-stateful\phase1-mockup> curl http://localhost:3000/info


StatusCode        : 200
StatusDescription : OK
Content           : {"status":"success","message":"Stateless server
                    information","data":{"server":"Stateless Mock Server v1.0","t
                    imestamp":"2026-03-04T12:31:18.975Z","requestCount":1,"client
                    Id":"anonymous","randomValue":...
RawContent        : HTTP/1.1 200 OK
                    Content-Security-Policy: default-src 'self';base-uri
                    'self';font-src 'self' https: data:;form-action
                    'self';frame-ancestors 'self';img-src 'self'
                    data:;object-src 'none';script-src 's...
Forms             : {}
Headers           : {[Content-Security-Policy, default-src 'self';base-uri
                    'self';font-src 'self' https: data:;form-action
                    'self';frame-ancestors 'self';img-src 'self'
                    data:;object-src 'none';script-src 'self';script-src-attr
                    'none';style-src 'self' https:
                    'unsafe-inline';upgrade-insecure-requests],
                    [Cross-Origin-Opener-Policy, same-origin],
                    [Cross-Origin-Resource-Policy, same-origin],
                    [Origin-Agent-Cluster, ?1]...}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 423
```

→ Stateless

From ClientA:

```shell
curl -X POST http://192.168.10.10:3001/session
```

Store the returned `Session-ID`, then:

```shell
curl http://192.168.10.10:3001/data -H "Session-ID: <id>"
```

🔵 REMOTE TESTS (Via Simulated Internet)

From ClientA:

```
curl http://10.10.12.2:3000/info
```

That will target **ServerB stateless** — NAT + routing should deliver

---

Session Test:

```
curl -X POST http://10.10.12.2:3001/session
```

→ ServerB returns a session ID

Then:

```
curl http://10.10.12.2:3001/data -H "Session-ID: <id>"
```

Check session persists across NAT

⚠️ TROUBLESHOOTING

❗ Cannot Reach Remote

- Gateway misconfiguration
- NAT ACL incorrect
- Wrong default gateway on clients

Check:

```
None
show ip route
ping 10.10.12.2 from R1
```

❗ Sessions Lose State

- Missing header
- Wrong container port
- Firewall blocking port 3001

🧠 ENGINEERING REFLECTION

1. **Stateless services don't depend on prior data**
2. **Stateful services require session headers**
3. NAT shouldn't affect logical session identity

📄 LAB README.md (for repository)

```
None
# Lab 4 — Stateful vs Stateless Across Public & Private Subnets

## Architecture
Two LANs connected via a simulated internet (10.10.12.0/30)
- LAN A: 192.168.10.0/24
- LAN B: 192.168.20.0/24
Routers provide static routing + NAT.

## Steps
```

```
### 1. Setup Network
Configure R1 and R2:
- Static routes
- NAT overload
- Verify ping

### 2. Deploy IaC
Configure `ServerA.env` and `ServerB.env`
Deploy via VS Code, PowerShell or Bash using:
```

python manage.py deploy --profile week02-stateless-stateful

```
None

### 3. Validate
- Local stateless vs stateful
- Remote stateless vs stateful
- Test session persistence
- Capture NAT translations

## Troubleshooting
- Check interface status, NAT ACLs
- Verify client gateways

## Extensions
- OSPF instead of static routing
- Load balancer in front of APIs
```